

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

SibSp \	Name	Sex	Age
0	Braund, Mr. Owen Harris	male	22.0
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1	Heikkinen, Miss. Laina	female	26.0
2	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
0	Allen, Mr. William Henry	male	35.0

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
In [9]: #Preprocess the dataset
print("Missing values in the dataset: \n")
print(df.isnull().sum())

#Handle missing data
df["Age"] = df["Age"].fillna(df["Age"].median())
print("\n")
print("Missing values in the dataset after handling AGE column: \n")
print(df.isnull().sum())
```

Missing values in the dataset:

```
PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked          2
dtype: int64
```

Missing values in the dataset after handling AGE column:

```
PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              0
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked          2
dtype: int64
```

```
In [11]: # Feature Selection
from sklearn.model_selection import train_test_split
X = df[["Age", "Pclass", "SibSp", "Parch"]]
y = df["Fare"]
#Splitting the data into training and testing
X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size=0.2,random_state=42)
print("DONE")
```

DONE

```
In [13]: #Linear regression Model
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train,y_train)
print("Model trained")
```

Model trained

```
In [14]: #evaluate model
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("MAE:", mae)
print("MSE:", mse)
print("R2 Score:", r2)
```

MAE: 19.781969090439063
MSE: 946.3355043476649
R2 Score: 0.3884468201561928

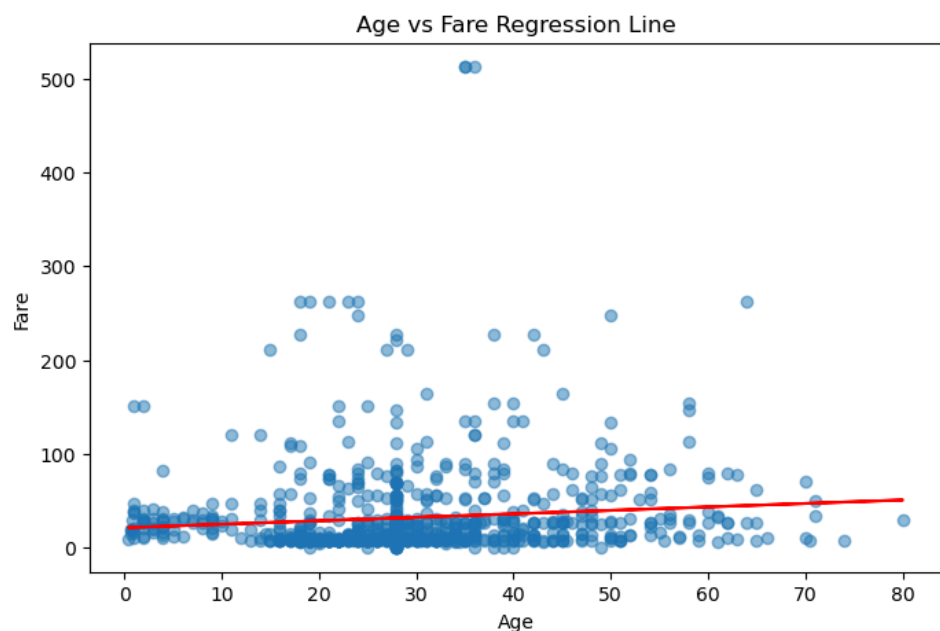
```
In [15]: #Plot linear regression
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

X = df[["Age"]]
y = df["Fare"]

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

plt.figure(figsize=(8,5))
plt.scatter(X, y, alpha=0.5)
plt.plot(X, y_pred, color="red")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Age vs Fare Regression Line")
plt.show()
```



```
In [16]: print("Intercept:", model.intercept_)
print("Coefficient:", model.coef_[0])

Intercept: 21.368631241135958
Coefficient: 0.36903926178178814
```

```
In [ ]:
```